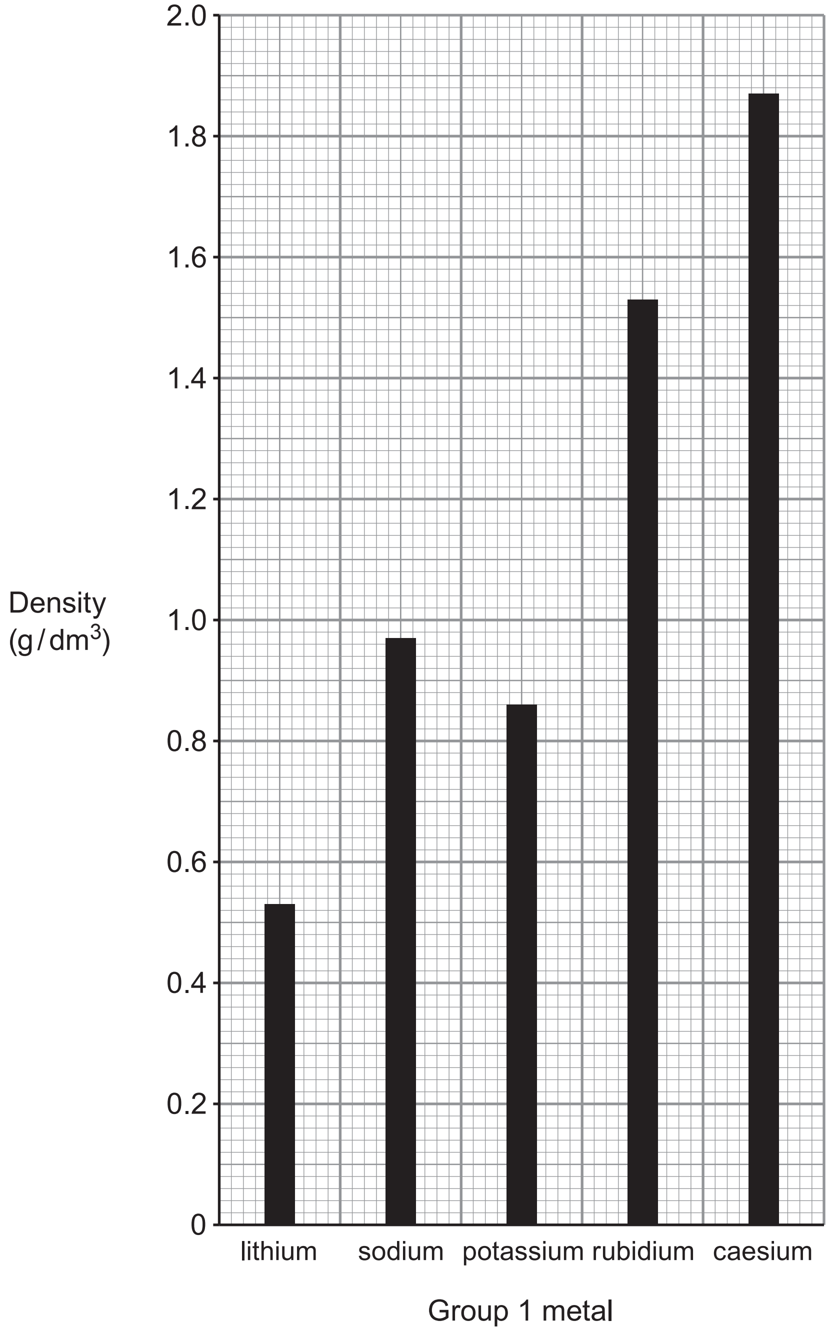
Examiner only

6. *(a)* The bar graph shows the densities of five Group 1 metals.



Draw a **straight** line on the bar graph showing the general trend in density of Group 1 metals. How does your trend line suggest that potassium and **not** sodium is the anomalous density value? [2]

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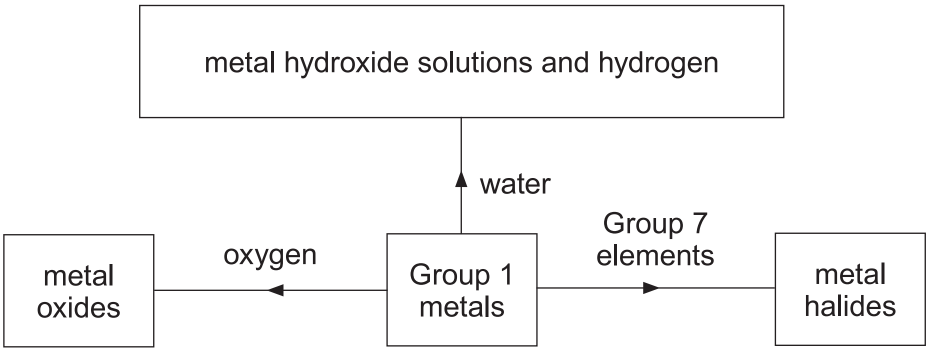
(b) The flow chart shows some reactions of Group 1 metals.



(i) State and explain **one** similarity and **one** difference you would see when lithium and potassium are added separately to a trough of cold water. [4]

Similarity ....

....

Difference ....

....

(ii) Write a balanced **symbol** equation to show the reaction between lithium and oxygen. [3]

....

(iii)

Group 1	**Group 7**
lithium sodium potassium	chlorine bromine iodine

The boxes show some of the elements in Group 1 and Group 7. Explain, in terms of electronic structure, why potassium and chlorine would react the most violently. [3]

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6. Limestone is an important raw material. It can be used as a building material or converted into quicklime and slaked lime.

Describe and explain the sequence of reactions carried out in the laboratory to convert limestone into slaked lime. [6 QER]

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6

6. This question is about the uses of gases present in air.

Air is a mixture of different gases. Each gas has different uses. The gases are present in different amounts as shown below.

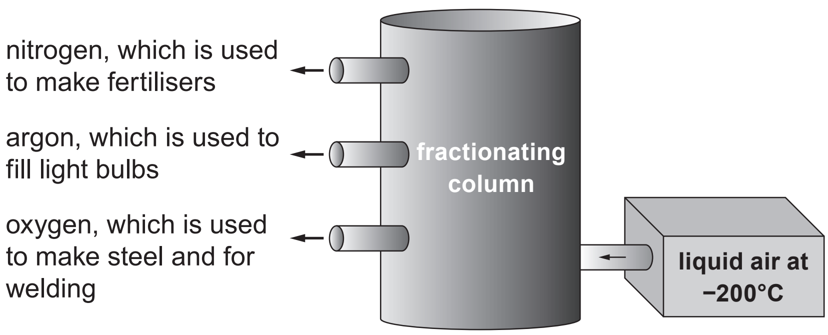
Gas	Amount present in air	Boiling point (°C)
N_2	78.04 %	−195.8
O_2	20.95 %	−183.0
CO_2	0.03 %	−78.5
Ar	0.93 %	−189.2
Ne	18 ppm	−246.0
He	5 ppm	−268.9
Kr	1 ppm	−152.3
Xe	0.08 ppm	−107.1
H_2	0.5 ppm	−257.9
CH_4	2 ppm	−164.0
N_2O	0.5 ppm	−88.5

ppm = parts per million

Separating a complex mixture!

Air is firstly compressed and cooled which turns it into a liquid. Carbon dioxide freezes and is removed as solid dry ice. The rest of the mixture goes on to a fractionating column where it is slowly warmed allowing most of the substances to be separated.

Air is filtered and compressed → Compressed air is cooled to −200 °C turning most substances into liquids → Solid carbon dioxide is removed from liquid air → Liquid air undergoes fractional distillation →



Examiner only

(a) Tick (✓) the **main** reason why hydrogen is not separated during the fractional distillation of liquid air. [1]

hydrogen is a highly reactive gas ☐

only 0.5 ppm of hydrogen is present ☐

hydrogen does not become liquid on cooling to −200 °C ☐

hydrogen has a higher boiling point than helium ☐

(b) Tick (✓) the reason why carbon dioxide becomes a solid during the first part of the process. [1]

carbon dioxide has a boiling point above −200 °C ☐

carbon dioxide has a melting point above −200 °C ☐

carbon dioxide has a melting point below −200 °C ☐

carbon dioxide has a boiling point below −200 °C ☐

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(c) Describe, in terms of boiling points, how nitrogen, argon and oxygen are separated during fractional distillation. [3]

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(d) The noble gases are present in very small quantities in the air. The amount of each gas produced per year is shown in the table.

Noble gas	World production (tonnes per year)
helium	28 000
neon	1 000
argon	700 000
krypton	8
xenon	0.6

Use the data to calculate the number of tonnes of air needed to produce 700 000 tonnes of argon. Give your answer in **standard form**. [2]

Mass of air needed = ... tonnes

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Examiner only

6. (a) **S** and **T** are two metal halide compounds. The ions in each compound were identified by observations from a flame test and the addition of silver nitrate solution.

(i) Complete the table. [4]

Compound	Flame test colour	Symbol of ion	Observation on adding silver nitrate solution	Symbol of ion
S	brick red	....	yellow precipitate	....
T	....	Ba^{2+}	white precipitate	....

(ii) Suggest why a silver nitrate test would not be able to distinguish clearly which halide ions are present in a **mixture** of compounds **S** and **T**. [1]

....

....

(b) Write an **ionic** equation for the formation of the precipitate in the reaction between sodium chloride solution and silver nitrate solution.

Include state symbols. [3]

.... + ... $\longrightarrow$ ....

8

Examiner only

6. (a) The table shows the approximate percentages of some gases found in the Earth's **original** atmosphere and the Earth's atmosphere **today**.

Gas	Approximate percentage (%) of gases in the Earth's atmosphere	
	Original	**Today**
carbon dioxide	75	0.04
water vapour	25	variable, between 1-2
oxygen	0	21

Explain how the changes to these gases have taken place over geological time. [6 QER]

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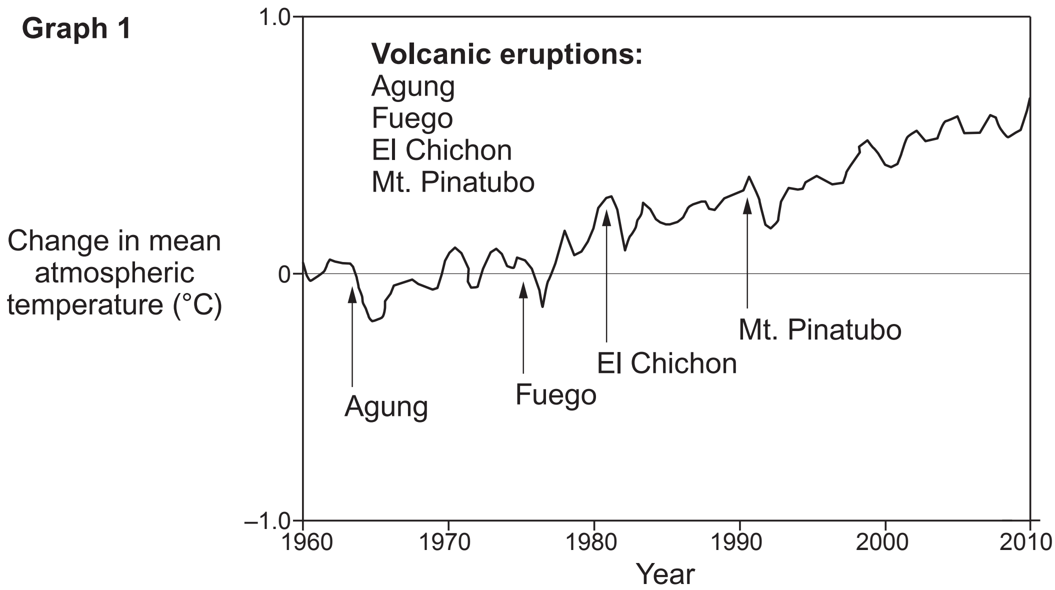
(b) Use the information on this page to answer parts (i)–(iii).

Do Volcanoes Cause Global Warming?

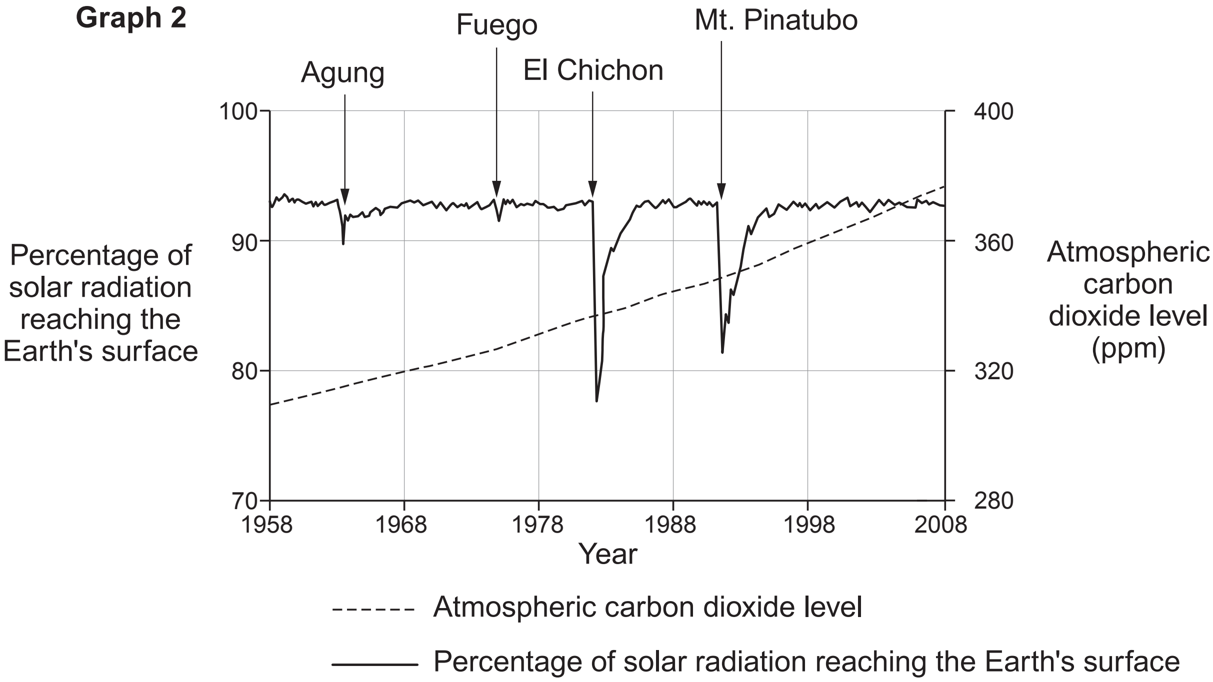
Volcanoes release vast amounts of greenhouse gases, such as carbon dioxide, which contribute to global warming. **Graph 1** shows the effect of four volcanic eruptions on the mean atmospheric temperature.

Erupting volcanoes also emit large quantities of sulfur dioxide into the atmosphere. Unlike carbon dioxide, sulfur dioxide increases the reflection of radiation from the Sun back into space, which cools the Earth's atmosphere. **Graph 2** shows the effect of the same four large volcanic eruptions on the percentage of solar radiation reaching the Earth's surface. The graph also shows the change in carbon dioxide levels in the atmosphere over the same time period.

Graph 1



Graph 2



Examiner only

(i) Tick (✓) the box next to the statement which best describes the effect of volcanic eruptions on the overall level of carbon dioxide in the atmosphere. [1]

No significant impact on the overall level of carbon dioxide ☐

A significant increase in the level of carbon dioxide ☐

A significant decrease in the level of carbon dioxide ☐

(ii) Tick (✓) the box next to the statement which best describes the effect that volcanic eruptions have on the mean atmospheric temperature. [1]

Mean atmospheric temperature decreases ☐

Mean atmospheric temperature increases ☐

No effect on the mean atmospheric temperature ☐

(iii) Tick (✓) the box next to the statement which best explains the change in solar radiation reaching the Earth's surface after volcanic eruptions. [1]

Solar radiation decreases because it is reflected by sulfur dioxide ☐

Solar radiation increases because it is absorbed by carbon dioxide ☐

Solar radiation increases because it is absorbed by carbon dioxide and sulfur dioxide ☐

Solar radiation decreases because it reacts with sulfur dioxide forming sulfuric acid ☐

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